

MGMT 590 Final Project — Report Outline & Writing Guide

Track: LLM Evaluation **Primary paper:** Huang, Burt, Hutter, Broderick (2025). *Dropping Just a Handful of Preferences Can Change Top Large Language Model Rankings*. arXiv:2508.11847
Secondary / context paper: Singh et al. (2025). *The Leaderboard Illusion*. arXiv:2504.20879
Target length: ≤ 8 pages (NeurIPS 2026 template) + references + appendix **Due:** May 1

This document is the group's shared blueprint for the final report. It explains the plan, the key terms, the content of each section, and who-does-what. Sections can be assigned independently.

0. Glossary — read this first

These abbreviations and technical terms show up throughout the outline and the report. Keep this glossary as a reference tab while drafting.

Term	Plain-English meaning
LLM	Large Language Model. GPT-5, Claude, Gemini, Llama, etc.
Arena-style evaluation	A way to compare LLMs by showing users two anonymous model outputs side-by-side and asking which is better. Votes are aggregated into a leaderboard. Chatbot Arena is the canonical example.
Pairwise comparison	A single "battle": two outputs, one winner. Preferred over absolute scoring because it's easier for humans.
Bradley–Terry model (BT)	The statistical model most leaderboards use to turn millions of pairwise battles into one number per model. Each model gets a latent "strength" score; the probability that model i beats j is a logistic function of their score difference. Dates to Bradley & Terry (1952).

Term	Plain-English meaning
Elo rating	The chess rating system. A presentation-layer rescaling of BT scores that Arena uses in its UI. Mathematically equivalent to BT under certain assumptions.
MLE (Maximum Likelihood Estimation)	The standard statistical procedure for fitting BT: find the strength scores that make the observed battle outcomes most probable.
Log-likelihood	The objective function MLE maximizes. Loosely: "how well does this set of scores explain the data we saw?"
Bootstrap confidence interval (CI)	A way to put error bars on an estimate by repeatedly resampling the data with replacement and refitting. Answers: "how much does the score wobble under random resampling?"
IID (Independent and Identically Distributed)	A strong statistical assumption that every data point is drawn from the same distribution and doesn't depend on other data points. Classical CIs assume IID; Arena votes probably aren't truly IID (same annotators, prompt-selection bias, etc.).
Influence function / influence score	A number attached to each individual data point measuring how much the final model estimate would change if that point were removed. Comes from classical robust statistics.
AMIP (Approximate Maximum Influence Perturbation)	A method (Broderick, Giordano, Meager, 2020) that uses influence scores to approximate the worst-case subset of data you could drop to flip a conclusion. The primary paper extends AMIP to BT-based rankings.
Worst-case data-dropping	A robustness check: instead of asking "what if data were resampled randomly?" it asks "what's the smallest evil subset whose removal changes the answer?"

Term	Plain-English meaning
Robustness	In this report: stability of the ranking when a small fraction of votes is removed.
α_{flip}	Our shorthand for the smallest fraction α of votes whose removal would flip a pairwise ranking decision. Small α_{flip} = fragile. Large α_{flip} = robust.
Goodhart's Law	"When a measure becomes a target, it ceases to be a good measure." Quoted by Singh et al. to explain why Chatbot Arena distorts over time.
MT-Bench	An older LLM benchmark that uses expert annotators and curated prompts. The primary paper finds it is more robust than crowdsourced Chatbot Arena.
LLM-as-Judge	Using a strong LLM (like GPT-4) to automatically rate outputs instead of human voters. Arena has a variant of this.
M-estimator	A generalization of MLE where you optimize a custom loss function. Robust M-estimators down-weight extreme data points.
Huber loss	A classic robust loss function: quadratic near zero, linear for large residuals. Huberizing the BT loss would cap how much any one vote can influence the fit.
Kendall-τ (tau)	A rank correlation metric. 1.0 = identical rankings; 0 = no relationship; -1 = reversed. Useful for comparing two leaderboard orderings.
RLHF / DPO	Reinforcement Learning from Human Feedback / Direct Preference Optimization. Training techniques that use the same kind of preference data Arena collects. Covered earlier in the course.

Term	Plain-English meaning
NeurIPS	Conference on Neural Information Processing Systems. The template the assignment requires. Two-column, 8 pages + refs + appendix.

Strategic framing (read second)

The report argues one unified thesis:

Arena-style LLM leaderboards suffer from **two coupled failure modes** — *system-level bias* (Singh et al., 2025) and *statistical fragility* (Huang et al., 2025). Current platforms report point rankings with bootstrap confidence intervals, but neither of those checks catches worst-case data-dropping instability, and neither addresses who submits data and how. We propose a **Robustness-Aware Leaderboard (RAL)** framework that makes fragility visible, actively repairs it by sampling influence-heavy matchups, and stress-tests ranking claims with influence-capped BT variants.

One paper deep, one paper as context, one concrete proposed solution. The report should resist becoming a survey.

Page budget

Section	Pages	Notes
Abstract	~0.2	Single-column abstract block
1. Introduction	~0.75	Motivation + contribution list
2. Paper Summary (assignment part a)	~1.5	Primary paper deep, secondary paper tight
3. New Research Direction (part b)	~1.0	Research question, gap, why it matters
4. Proposed Solution (part c)	~2.0	Three components + feasibility

Section	Pages	Notes
5. Literature Review (part d)	~1.5	Five clusters
6. Preliminary Evidence (part e, optional)	~0.75	Toy experiment on arena-human-preference-140k dataset
7. Discussion & Limitations	~0.3	
Total	~8	
References	—	Not counted toward 8 pages
Appendix	—	Not counted toward 8 pages

Title options (pick one as a team)

1. **Robustness-Aware LLM Leaderboards: From Diagnosing Ranking Fragility to Actively Repairing It**
2. Beyond the Confidence Interval: Influence-Guided Sampling and Robust Aggregation for Arena-Style LLM Evaluation
3. When One Vote Moves the Throne: A Robustness-First Framework for Preference-Based LLM Ranking

Recommended: #1. It pairs diagnosis with prescription, signals a concrete contribution, and reads well on a title slide.

Abstract (draft — target ~180 words)

Arena-style evaluation has become the default way the field compares frontier LLMs, but recent work shows the resulting rankings are simultaneously (i) structurally biased by who submits models and how data is distributed, and (ii) statistically fragile — Huang et al. (2025) report that removing roughly 0.003% of preferences can flip the top-ranked model on Chatbot Arena. Current platforms publish bootstrap confidence intervals, which quantify sampling variability but not worst-case data-dropping sensitivity. We propose a **Robustness-Aware Leaderboard (RAL)** framework with three components: (1) a **robustness interval** reported alongside each Bradley–Terry score, giving the minimum fraction of votes whose removal would flip a pairwise ranking; (2) **influence-guided active**

sampling, which re-weights the Arena's matchup selector toward pairs whose scores are most fragile; and (3) **influence-capped BT aggregation**, a robust M-estimator variant that caps the leverage of individual votes. We describe a feasible toy-experiment path using the public [lmarena-ai/arena-human-preference-140k](#) dataset and situate RAL within the growing literature on ranking reliability, vote manipulation, and low-rank evaluation.

1. Introduction

Opening hook (2–3 sentences)

Leaderboards now drive procurement, research direction, and billions in investment. When a single leaderboard position shifts hiring, funding, or product decisions, the robustness of that number matters. Recent work shows two independent reasons to doubt it.

Problem statement (1 paragraph)

Pairwise preference evaluation (Chatbot Arena; Chiang et al., 2024) aggregates millions of human votes via the Bradley–Terry (BT) model (Bradley & Terry, 1952) into a single leaderboard. Two complementary critiques have emerged:

- **System-level bias** — *The Leaderboard Illusion* (Singh et al., 2025) documents private testing, data-access asymmetries, and silent model deprecation.
- **Statistical fragility** — Huang et al. (2025) show rankings can flip after removing a fraction of preferences so small it is numerically imperceptible.

Gap (1 paragraph)

Existing confidence intervals on BT (produced by bootstrap resampling) quantify variability when the data are re-drawn at random. That is a different question from worst-case data-dropping robustness — the question of whether some *specific* small subset of votes is propping up the ranking. Huang et al. (2025) make this contrast explicit. No deployed leaderboard currently (a) reports robustness intervals, (b) uses robustness — rather than only information gain — to schedule new votes, or (c) uses aggregation methods designed to resist high-leverage observations.

Contributions (bullet list at end of intro)

1. We synthesize system-level and statistical failure modes of Arena-style evaluation into one conceptual frame.
2. We formulate a concrete research question about **robustness-aware ranking**.

3. We propose a three-component framework (Robustness-Aware Leaderboard, or RAL) that is both diagnostic and prescriptive.
 4. We outline a feasible empirical program using the public [arena-human-preference-140k](#) dataset.
 5. We map the proposal against the current state of the art in preference-based evaluation.
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2. Paper Summary (Assignment Part a)

The assignment asks for one paper, but the natural scientific unit here is two papers read in conversation. Section 2.1 is the deep dive; Section 2.2 is a tight contextual summary that motivates the research direction in Section 3.

2.1 Primary paper — Huang et al. (2025)

Main idea (1–2 sentences)

The paper develops a computationally cheap method to ask: *if an adversary (or natural noise) removed the worst-case small fraction of preference votes, would the leaderboard's top-ranked model change?* Applied to Chatbot Arena and derivative leaderboards, the answer is yes — at fractions as small as 0.003%.

Key findings worth citing in the report

- Roughly **0.003% of votes** is enough to flip the top-ranked model on Chatbot Arena (Human Judge version).
- **MT-Bench is notably more robust** (larger required drop fraction) than Chatbot Arena, which the authors attribute to expert annotators and curated prompts.
- Fragility **persists inside bootstrap confidence intervals** — meaning AMIP catches a failure mode that reported CIs do not.

(Paraphrase these in your prose. Do not string together multiple verbatim quotes from the paper.)

Methodology — plain-English walkthrough

What are they trying to estimate? The Bradley–Terry (BT) latent score for each model. Chatbot Arena fits BT using maximum likelihood on a giant pile of pairwise battle outcomes (Chiang et al., 2024).

What's the robustness question? For a given pairwise ranking decision — say, "is model A above model B?" — how many votes would need to be removed to reverse that decision?

Why not brute-force it? With 140k votes, checking every possible subset of size 500 (say) is astronomically many combinations. Computationally impossible.

The trick — influence scores. Each data point has an *influence score*: a Taylor-expansion approximation of how much the model fit would change if that point were removed. Points with extreme scores are "high-leverage." Influence scores can be computed in one pass after the model is fit.

AMIP (Approximate Maximum Influence Perturbation) in one sentence. Rank every vote by its influence score; greedily remove the top- α ones; predict whether this would flip the ranking; then actually refit the BT model on the remaining data to verify.

What they actually ran:

- AMIP (Broderick et al., 2020) on the BT log-likelihood.
- A refinement called Additive One-step Newton (Huang et al., 2025a) for more accurate approximation.
- Refitting BT after the predicted influential subset is removed, to confirm whether the predicted flip actually occurs.

Cost. Dominated by the initial BT fit. After that, influence scoring is $O(N)$. Easily runs on a laptop.

Datasets tested. Chatbot Arena (human vote and LLM-judge variants), MT-Bench, Vision Arena, Search Arena, and other derivative platforms.

Why AMIP is *different* from bootstrap confidence intervals (the crucial distinction)

This is the single most important technical point in the paper. Memorize this contrast.

- **Bootstrap CI** asks: "*If I redrew votes randomly from the population, how much would the BT score wobble?*" Assumes the data are roughly IID.
- **AMIP** asks: "*Is there any small subset I could delete that would reverse the conclusion?*" Makes no IID assumption.
- A result can be **statistically significant** (tight CI, doesn't cross zero) and still **AMIP-non-robust** (a handful of votes are propping it up). Broderick et al. (2020) first showed this for econometrics; Huang et al. (2025) show it for LLM rankings.

In short: CIs and AMIP measure *different* kinds of uncertainty. Reporting only one misses the other.

Key contribution / why it matters

- First robustness audit of BT-based LLM rankings.

- Interpretable output: AMIP identifies the *specific votes* responsible for a flip, enabling manual inspection of whether those votes are noise, mistakes, or legitimate preferences.
- Crowdsourced systems (Chatbot Arena) are less robust than expert-curated systems (MT-Bench). Suggests that *data quality* matters more than *data quantity* for rank stability.

Honest limitations (include these to show critical reading)

- AMIP is an approximation. Follow-up work by the same group (Huang, Burt, Shen, Nguyen, Broderick, 2024/2025 — *Approximations to worst-case data dropping: unmasking failure modes*) shows AMIP can underestimate the true worst-case drop in some configurations.
- "Non-robust" does not mean "wrong." It means the ranking is *contingent on a small subset* of the data — which is unsettling, but not a proof of error.
- The paper is diagnostic. It audits; it does not prescribe a fix. That gap is exactly what our proposed framework (Section 4) fills.

2.2 Secondary / context paper — Singh et al. (2025)

Keep this to roughly three paragraphs. Its job is to motivate *why* fragility matters: a fragile ranking living on top of a biased platform compounds into an illusion of progress.

Scope

An audit of approximately 2 million battles, 243 models, and 42 providers on Chatbot Arena, covering both public and private variants.

Four systemic failures the paper documents

1. **Private testing with selective disclosure.** Major providers can test multiple private variants and only release the best. The extreme case: 27 Meta variants tested ahead of the Llama-4 launch.
2. **Data-access asymmetry.** Roughly 19.2% of all Arena data goes to Google, 20.4% to OpenAI; the 83 open-weight models in the study collectively receive about 29.7%.
3. **Benchmark overfitting.** Even limited additional Arena data can produce relative performance gains of up to 112% on ArenaHard, a benchmark drawn from the Arena distribution. Providers who have more Arena data can tune harder to the benchmark without improving on general tasks.
4. **Opaque model deprecation.** 205 models were silently removed from the leaderboard without notification, breaking the consistency of the comparison set.

Goodhart framing

Singh et al. invoke **Goodhart's Law** — when a measure becomes a target, it ceases to be a good measure. This is the bridge to our framing: fragility is not only a statistics problem, it is a *governance* problem.

Bridge sentence to Section 3 (use in the paper)

System-level bias and statistical fragility are not independent problems: selective reporting introduces a small number of extreme data points, which are precisely the observations AMIP flags as high-leverage. Robustness-aware evaluation therefore speaks to both critiques at once.

3. New Research Direction (Assignment Part b)

3.1 Research question (state in one sentence, bold)

RQ. How can preference-based LLM leaderboards be augmented so that ranking fragility is made *visible*, *repairable*, and *resistant* to worst-case data perturbations — without sacrificing the scale and openness that make Arena-style evaluation useful?

3.2 Sub-questions (three, maximum)

1. **Diagnostic.** What is the right object to report alongside a BT score so that readers can tell whether a rank is robust? (The analogy: confidence intervals quantify sampling uncertainty; we want an analog for drop-robustness.)
2. **Prescriptive.** Can the Arena's active matchup selector be re-weighted to target fragile pairs — turning influence scores into a signal for *collecting* new data, not just auditing old data?
3. **Robust estimation.** Are there drop-in replacements for vanilla BT that are less sensitive to high-leverage preferences, while preserving interpretability and scalability?
4. **Heterogeneity.** How does judge heterogeneity affect the robustness and validity of preference-based rankings? Recent work shows that treating all judges equally can lead to biased rankings and misleading uncertainty estimates, even as data grows - more data can make evaluation more confidently wrong.

3.3 Why this matters in practice (the "so what")

- **Decision quality.** When procurement, model-routing, and research priorities are pinned to a leaderboard rank, knowing whether that rank would survive a handful of different votes is a first-order business question.
- **Trust and reproducibility.** Robustness intervals give users a principled way to read "Model A: rank 2 (robust)" vs. "Model A: rank 2 (fragile to ~12 votes)" without forcing them to re-run an AMIP audit themselves.
- **Incentive alignment.** Under the system-level bias lens (Singh et al., 2025), robust aggregation reduces the marginal value of submitting extra variants because high-leverage votes are down-weighted. This is a structural, not just statistical, benefit.
- **Generality.** Our method extends to any BT-based ranking: RewardBench, LLM-as-Judge leaderboards, and the preference datasets used in RLHF and DPO. Contributions travel.
- **Tensor Completion.** Recent work suggests that LLM evaluation can be framed as a low-rank tensor completion problem under highly non-uniform sampling, highlighting that current leaderboard estimates fail to exploit shared structure across tasks and contexts

3.4 Scope and non-goals (be explicit)

- **In scope:** BT-based rankings; crowdsourced and expert-curated preference data; an offline re-analysis; a proposed online sampling rule.
- **Out of scope:** detecting adversarial vote rigging (Huang et al., 2025b; Min et al., 2025 — referenced but not solved here); fundamentally new judge architectures; multi-dimensional preferences beyond a scalar BT score.

4. Proposed Solution (Assignment Part c)

This is the core of the report. Give it roughly two pages. Structure as three components that fit together, not three disconnected methods.

4.1 The Robustness-Aware Leaderboard (RAL) — overview figure

The report's "hero figure" should show a leaderboard row with three numbers per model instead of one. Example:

Model	BT score	95% CI	α_{flip} (min drop to lose rank)
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GPT-5	1412	[1405, 1419]	0.0031% ← fragile
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Claude-4.6 1407 [1399, 1414] 0.41% ← moderately robust

Gemini-3 1398 [1389, 1407] 2.8% ← robust

The α_{flip} column is the operational output of Component 1. This picture is the clearest way to communicate the entire contribution on a single slide.

4.2 Component 1 — Robustness intervals (diagnostic)

Current robustness metrics assume homogeneous votes, ignoring judge reliability differences. A natural extension is to integrate judge-aware weighting into influence estimation, so that high-influence votes from unreliable judges are down-weighted. A natural extension is to integrate judge-aware weighting into influence estimation, so that high-influence votes from unreliable judges are down-weighted.

Goal. For each pairwise ranking decision (i, j) , report $\alpha_{\text{flip}}(i, j)$: the approximate minimum fraction of votes whose removal would flip the sign of $(\text{score}_i - \text{score}_j)$.

Method.

1. Fit BT on the full Arena dataset (MLE, the standard approach used by Chatbot Arena).
2. Compute per-vote influence scores using AMIP (Broderick et al., 2020), applied to the score-difference target.
3. Sort influence scores; find the smallest α such that removing the top- αN influential votes predicts a sign flip.
4. Report α_{flip} per adjacent ranked pair (at minimum, between consecutive models in the leaderboard).

Cost. Dominated by a single BT fit; influence computation is $O(N)$. Already demonstrated feasible on full Chatbot Arena by Huang et al. (2025).

Novelty vs. the primary paper.

- Huang et al. (2025) run AMIP as a one-off audit.
- We propose it as a *recurring platform-level report* — a new column on the public leaderboard, updated whenever BT is refit.
- We extend from "top-1 flip" (their focus) to *pairwise* robustness across all adjacent ranking decisions.

4.3 Component 2 — Influence-guided active sampling (prescriptive)

Recent work shows that evaluation data exhibit low-rank structure across models and tasks , suggesting that sampling strategies should exploit shared structure rather than treating each pair independently. This implies that influence-based sampling could be combined with low-rank uncertainty estimates to prioritize structurally informative comparisons.

Goal. Given a budget of N new human votes, allocate them to reduce ranking *fragility* rather than only score *variance*.

Background. Chatbot Arena already uses active sampling for information gain — it prioritizes battles that will tighten confidence intervals (Chiang et al., 2024). Inclusion Arena uses a related strategy called Proximity Sampling that favors close-score pairs. Neither method explicitly targets fragile rankings.

Our proposal — influence-gain sampling. Schedule the next matchup (i, j) with probability proportional to the expected *reduction in influence* of existing votes on the ranking decision between i and j . Intuition: spend votes where the current ranking rests on a thin, fragile slice of history.

Formally, approximate influence gain as a function of:

- Current $\alpha_{\text{flip}}(i, j)$.
- The number of existing battles between i and j .
- Predicted battle outcome distribution under the current BT fit.

Feasibility. Slots into the existing matchup selector as an additional term; no new human infrastructure needed; uses votes the Arena is already collecting.

Novelty. Huang et al. (2025) diagnose fragility. We close the loop: we use the diagnosis to change what data gets collected next. This moves the field from post-hoc auditing toward self-correcting leaderboards.

Honest open question. Influence-gain sampling could bias future rankings *toward* currently-fragile pairs, potentially slowing convergence elsewhere. Needs a mixing parameter λ that balances information-gain and influence-gain objectives. Worth flagging in Section 7.

4.4 Component 3 — Influence-capped BT aggregation (robust estimation)

Instead of assuming homogeneous evaluators, we can extend BT to incorporate judge-specific reliability parameters, which has been shown to improve ranking accuracy and uncertainty calibration. This suggests a robust + judge-aware BT model.

Goal. Prevent any single vote from exerting disproportionate influence on BT scores.

Candidate estimators (pick one as primary, mention others as alternatives):

- **Primary: weighted BT with influence-based weights.** After an initial BT fit, compute each vote's influence score; down-weight votes whose influence magnitude exceeds a threshold (for example, cap the top 0.1% by $|\text{influence}|$); refit.
- **Alternative 1: Huberized BT loss.** Replace the log-likelihood with a Huber-robust version of the logistic loss. This is a textbook robust-statistics move — quadratic near zero, linear in the tails — and is a small optimization change.
- **Alternative 2: Rank-based aggregation as a sanity check.** Compare BT rankings to rankings produced by Plackett–Luce or Copeland-score methods. Disagreements flag pairs where BT's assumptions are doing the work.

Interaction with Components 1 and 2.

- Component 3 reduces the number of pairs Component 1 flags as fragile.
- Component 2's sampling signal should be computed *after* Component 3's re-weighting, so the whole system is internally consistent.

Feasibility. All three estimators are MLE or M-estimator variants — a change in the optimizer's loss function, not an architectural overhaul. The Chatbot Arena team has already open-sourced the BT fitting pipeline (LMSYS blog, 2023); swapping in a Huber loss is realistically a day of engineering.

4.5 Evaluation plan — what "solved" looks like

A successful RAL should satisfy three claims, evaluated on public Arena data:

Claim	How to test
C1. RAL's α_{flip} intervals are well-calibrated: a pair reported as robust should rarely flip under held-out vote subsets.	Split the 140k dataset into train / held-out; predict α_{flip} on train; verify on held-out.

Claim	How to test
C2. Influence-gain sampling reaches a target α_{flip} threshold with fewer new votes than information-gain sampling.	Simulate both samplers on the 140k dataset using leave-out votes as synthetic "new" votes. Compare the number of votes needed to reach a target robustness level.
C3. Influence-capped BT produces rankings more stable under AMIP audit than vanilla BT, at negligible cost in held-out predictive log-likelihood.	Refit both on 140k; compare held-out log-likelihood and α_{flip} distributions.

5. Literature Review (Assignment Part d)

Five clusters. One short paragraph (3–5 sentences) each. This is the second-most-scrutinized section after the proposed solution.

5.1 Preference-based LLM evaluation and the Bradley–Terry backbone

- Chatbot Arena formalized crowdsourced pairwise evaluation at scale and adopted BT as its ranking backbone (Chiang et al., 2024; Zheng et al., 2023; LMSYS blog, 2023).
- BT itself dates to Bradley & Terry (1952). Hunter (2004) gives the standard MM-algorithm estimator (a specific way to maximize the BT likelihood).
- Inclusion Arena (Zhou et al., 2025) extends the paradigm to real-world app data with Proximity Sampling.
- **Gap:** all of this work assumes the BT score is the correct object to report. None of it provides drop-robustness guarantees.

5.2 Critiques of Arena-style leaderboards

- **The Leaderboard Illusion** (Singh et al., 2025) — private testing, data-access asymmetries, overfitting, silent deprecation.
- **Vote rigging** (Huang et al., 2025b; Min et al., 2025) — shows that hundreds of rigged votes can promote a target model by multiple ranks by exploiting BT's interconnected score structure.
- **Gap:** these papers establish *why* the current leaderboard system has exploitable weak points but do not propose statistical estimators that resist those weak points.

5.3 Worst-case data-dropping and robustness metrics

- **AMIP** (Broderick, Giordano, Meager, 2020) — originates the framework; applied to economics, not rankings.
- Follow-ups on approximation quality: Kuschnig et al. (2021); Moitra & Rohatgi (2023); Freund & Hopkins (2023); Nguyen et al. (2024); Rubinstein & Hopkins (2025); Huang, Burt, Shen, Nguyen, Broderick (2024/2025). Some of this work shows AMIP can miss true worst-case drops in adversarial configurations.
- **Huang, Burt, Hutter, Broderick (2025)** — our primary paper; first application to BT-based rankings.
- **Shiffman et al. (2023)** — the only prior ranking-robustness work we found, but it studies gene-set enrichment p-values, not preference ranking.

5.4 Uncertainty quantification for LLM rankings

- Confidence intervals via bootstrap on BT (Chiang et al., 2024; LMSYS blog, 2023).
- *Quantifying Uncertainty in LLM Benchmark Scores* (arXiv:2410.03492) — broader uncertainty quantification for benchmarks.
- *A Judge-Aware Ranking Framework* (arXiv:2601.21817 from the assignment reading list) — ranking with uncertainty over judges.
- **Gap:** all of this is sampling-variability uncertainty. None of it is worst-case-drop uncertainty.
- Recent work highlights that judge heterogeneity is a fundamental source of bias in LLM evaluation. Xu et al. (2026) propose a judge-aware extension of the Bradley–Terry model that jointly estimates model quality and judge reliability, showing that ignoring judge differences can produce biased rankings and invalid uncertainty estimates

5.5 Extensions and alternatives to BT

- **Prompt-to-Leaderboard (P2L)** (Frick et al., 2025, arXiv:2502.14855) — trains a small model to predict BT coefficients *conditional on a prompt*, producing prompt-specific leaderboards. Addresses the aggregation limitation but not robustness.
- *A Statistical Framework for Ranking LLM-Based Chatbots* (arXiv:2412.18407) — Rao–Kupper and Davidson generalizations of BT that handle ties more rigorously.
- *LLM Evaluation as Tensor Completion* (arXiv:2604.05460) and *Rich Insights from Cheap Signals* (arXiv:2603.02029) — low-rank structure for efficient evaluation.
- *Rethinking Bradley–Terry Models in Preference-Based Reward Modeling* (arXiv:2411.04991) — questions BT's underlying assumptions for reward modeling.
- *Fewer Battles, More Gain* (OpenReview XUVqFRp9oi) — Fisher-information-based active sampling. This is the closest prior work to our Component 2 in spirit, but it optimizes variance, not drop-robustness.
- Li et al. (2026) formulate LLM evaluation as a low-rank tensor completion problem, demonstrating that leaderboard estimation can be viewed as semiparametric inference

under highly non-uniform sampling. However, this line of work focuses on statistical efficiency rather than robustness to worst-case perturbations.

5.6 Where RAL sits in the literature (summary paragraph)

Robustness-Aware Leaderboards extend three lines of work at once: they apply AMIP-style drop-robustness (Broderick et al., 2020; Huang et al., 2025) as a platform-level report rather than a one-off audit; they generalize active sampling (Chiang et al., 2024; *Fewer Battles, More Gain*) beyond information gain to influence gain; and they replace vanilla BT MLE with a robust M-estimator while preserving the Chatbot Arena reporting pipeline. To our knowledge, no prior work combines these three into a single deployable framework, and none has been evaluated on the public [arena-human-preference-140k](#) release.

6. Preliminary Evidence (Assignment Part e — optional but encouraged)

The assignment explicitly points at

[lmarena-ai/arena-human-preference-140k](#). Even one mini-experiment produces a defensible Figure 1 and strengthens the report significantly. Each experiment below is achievable in Python with pandas + scipy + statsmodels in under a day.

6.1 Mini-experiment 1 — Replicate the core Huang et al. finding on 140k

- **Setup.** Fit BT on the full 140k dataset. For ties, follow Chiang et al. (2024) and count each as half a win and half a loss.
- **Compute.** AMIP influence scores for the top 5 pairwise ranking decisions. Estimate α_{flip} for each pair.
- **Deliverable.** A replication table showing α_{flip} for the top of the leaderboard. If we reproduce the "sub-1% flips top" finding, we have external validity for our framework.

6.2 Mini-experiment 2 — Vanilla BT vs. influence-capped BT

- **Setup.** Fit both vanilla BT and an influence-capped BT (cap the top 0.1% of votes by influence magnitude).
- **Compare.**
 - Held-out predictive log-likelihood on a random 20% split.
 - α_{flip} for the top-3 pairs under each estimator.
 - Kendall- τ rank correlation between the two leaderboards.

- **Deliverable.** Evidence for Claim C3 — similar predictive power, meaningfully higher α_{flip} .

6.3 Mini-experiment 3 — Simulated influence-gain sampler

- **Setup.** Hold out 10% of votes. Assume a vote budget of 500. Compare three samplers:
 - Uniform (baseline).
 - Information-gain (Fisher-information-based, current Arena-style approach).
 - Influence-gain (our proposal).
- **Metric.** Curve of α_{flip} vs. the number of added votes for the current top-3 pairs.
- **Deliverable.** Either we beat the baseline (supports Claim C2) or we don't (reporting honest ablation data is still a contribution).

6.4 If the group only has time for one experiment

Run **Mini-experiment 1**. It demonstrates that the problem our framework is designed to fix is real on the specific dataset the course provided. Use the α_{flip} bar chart across the top-10 pairs as Figure 1 of the report.

6.5 Compute and time estimates

- BT fit on 140k votes: minutes. Under the hood it's logistic regression on a design matrix of model indicators.
- AMIP influence scores: $O(N \cdot d)$, where d is the number of models. On 140k votes with roughly 100 models, under a minute on a laptop after the BT fit.
- All three experiments together: realistically 4–8 focused hours once a data loader is written.

6.6 Division of labor (suggested)

Task	Best fit for
Data loader for 140k dataset (pandas + tie handling)	Person comfortable with data wrangling
BT fit + influence-score computation	Person comfortable with statsmodels / scipy
Figure production (α_{flip} bar chart, Kendall- τ scatter)	Person comfortable with matplotlib
Writing Sections 1, 2, 3	Strongest writer
Writing Section 4	Person most comfortable with the method

Task	Best fit for
Writing Section 5	Person most comfortable with fast literature triage

7. Discussion, Limitations, Open Problems

Our framework assumes homogeneous judges and scalar latent scores. However, recent work suggests that both judge heterogeneity and low-rank structure across tasks play a critical role in evaluation reliability. Incorporating these dimensions into robustness-aware evaluation remains an important direction for future work.

Limitations to name explicitly

1. **AMIP is an approximation.** Huang, Burt, Shen, Nguyen, Broderick (2024/2025) show configurations in which AMIP underestimates worst-case drops. Our α_{flip} is therefore a *lower bound on fragility* — the true worst-case may be smaller.
2. **Influence-gain sampling could create its own pathologies.** If providers learn that fragile pairs receive more votes, they may design outputs to stay in that regime. Worth flagging as future work.
3. **Influence-capped BT changes the estimand.** Downstream consumers who assume vanilla BT semantics (including the Elo-like interpretation) will need to recalibrate.
4. **We do not address judge noise.** LLM-as-Judge bias (*Judging the Judges*, arXiv:2406.07791) is orthogonal; our framework assumes the judgments are given, whatever they are.
5. **The Singh et al. system-level critique is not fully resolved.** Data-access asymmetry is a governance problem, not a purely statistical one. RAL *reduces* its leverage; it does not eliminate it.

Future work

- **Prompt-level RAL:** combine RAL with Prompt-to-Leaderboard (Frick et al., 2025) to produce prompt-conditional robustness intervals.
 - **RAL under adversarial rigging:** compare Component 3 against the T-Normal and Omni-BT attacks of Huang et al. (2025b).
 - **Human-in-the-loop audit:** present flagged high-influence votes to annotators for re-adjudication.
-

8. Quick-reference cheat-sheet

Primary numbers to cite

- 0.003% of votes — enough to flip the top-ranked Chatbot Arena model (Huang et al., 2025).
- 19.2% / 20.4% / 29.7% — data split among Google / OpenAI / 83 open-weight models (Singh et al., 2025).
- 27 — Meta private variants tested before the Llama-4 launch (Singh et al., 2025).
- ~112% — relative performance gain on ArenaHard from additional Arena data (Singh et al., 2025).
- 205 — models silently deprecated (Singh et al., 2025).

Named methods we build on

- BT model (Bradley & Terry, 1952).
- AMIP (Broderick et al., 2020).
- Additive One-step Newton (Huang et al., 2025).

Named methods we extend past

- Bootstrap CIs on BT (Chiang et al., 2024).
- Information-gain active sampling (Chiang et al., 2024; *Fewer Battles, More Gain*).
- P2L (Frick et al., 2025).

Named datasets

- [lmarena-ai/arena-human-preference-140k](#) — the dataset the assignment points to.
- [lmsys/chatbot_arena_conversations](#) — 33k conversations with pairwise preferences, released 2023.

The single most important conceptual contrast in the report (commit to memory)

- **Bootstrap CI** = variability under random resampling; assumes IID data.
- **AMIP** = worst-case change under a small data drop; no IID assumption.
- They measure fundamentally different things. Reporting only one misses the other.